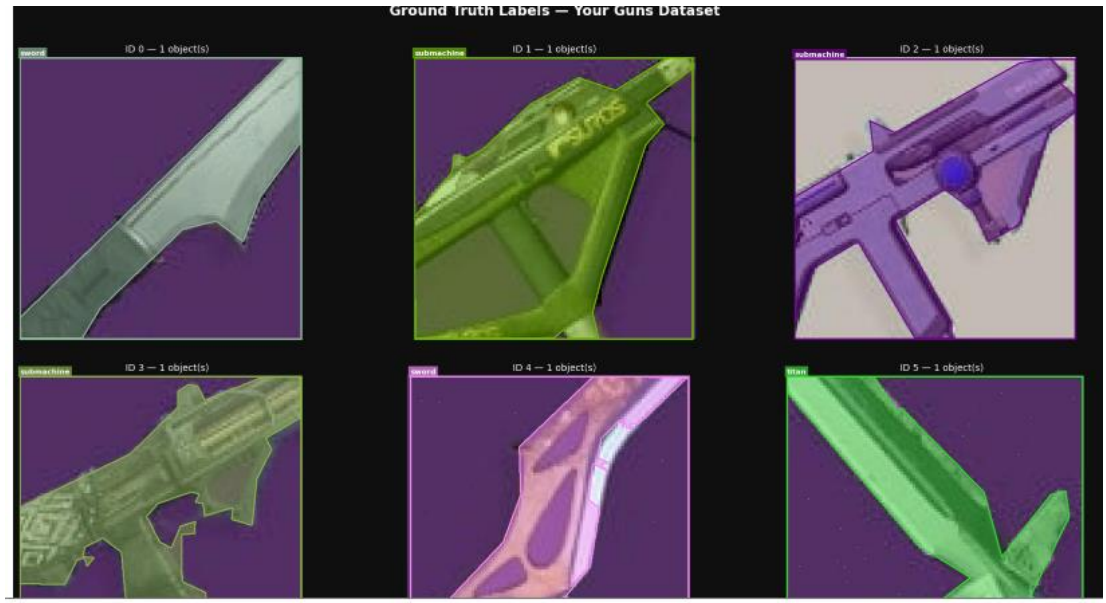
Tutorial 11
Instance Segmentation

Task 1: Dataset Preparation

- Dataset: Custom Guns dataset in COCO format.
- Training images: 345 ,Validation images: 14 , Test images: 15
- Training annotations: 369
- Number of classes: 8

```
📊 DATASET STATISTICS
────────────────────────────────────────
Training images     : 345
Training annotations: 369
Number of classes   : 8

Classes found:
  • [ 0] Guns                  –   0 instances
  • [ 1] auto_riff             –   3 instances
  • [ 2] snipper               –  15 instances
  • [ 3] submachine            – 228 instances
  • [ 4] sword                 –  87 instances
  • [ 5] titan                 –   9 instances
  • [ 6] trace riffle          –  24 instances
  • [ 7] warlock               –   3 instances
```



Task 2: Model Setup

- Train : 345 images (173 batches)
- Valid : 14 images (14 batches)
- Test : 15 images (15 batches)
- Batch size : 2

Mask R-CNN Configuration

- Output classes : 9 (background + 8 weapon classes)
- Trainable params : 43,737,669
- Optimizer : SGD lr=0.005 . StepLR step=3 gamma=0.1
- Device : cuda
- Epochs : 10

```
STEP 5 COMPLETE
Train : 345 images
Valid : 14 images
Test  : 15  images
Classes (9): ['background', 'Guns', 'auto_riff', 'snipper', 'submachine', 'sword', 'titan', 'trace riffle', 'warlock']

STEP 6 COMPLETE — DataLoaders ready
Train batches : 173
Valid batches : 14
Test batches  : 15

STEP 7 — Building Mask R-CNN...

STEP 7 COMPLETE
Output classes    : 9  →  ['background', 'Guns', 'auto_riff', 'snipper', 'submachine', 'sword', 'titan', 'trace riffle', 'warlock']
Trainable params  : 43,737,669

STEP 8 COMPLETE — SGD lr=0.005 | StepLR step=3 gamma=0.1

============================================================
STARTING TRAINING
Epochs     : 10
Batch size : 2
Device     : cuda
Classes    : ['background', 'Guns', 'auto_riff', 'snipper', 'submachine', 'sword', 'titan', 'trace riffle', 'warlock']
============================================================
```

Task 3 : Training Log

```
[  1/173] total=3.6927 | cls=2.0113 | box=0.1026 | mask=1.5535 | rpn=0.0086 (0.6s)
[  6/173] total=0.8641 | cls=0.1400 | box=0.1000 | mask=0.6007 | rpn=0.0114 (2.8s)
[ 11/173] total=0.6862 | cls=0.0867 | box=0.0781 | mask=0.5012 | rpn=0.0110 (5.0s)
[ 16/173] total=0.5150 | cls=0.0703 | box=0.1067 | mask=0.3188 | rpn=0.0026 (7.3s)
[ 21/173] total=0.3612 | cls=0.0626 | box=0.0921 | mask=0.1722 | rpn=0.0244 (9.5s)
[ 26/173] total=0.4554 | cls=0.0599 | box=0.0623 | mask=0.3123 | rpn=0.0060 (11.8s)
[ 31/173] total=0.5472 | cls=0.0493 | box=0.0808 | mask=0.3946 | rpn=0.0052 (14.1s)
[ 36/173] total=0.3468 | cls=0.0424 | box=0.0494 | mask=0.2360 | rpn=0.0047 (16.8s)
[ 41/173] total=0.3033 | cls=0.0374 | box=0.0763 | mask=0.1728 | rpn=0.0067 (19.2s)
[ 46/173] total=0.2941 | cls=0.0487 | box=0.0742 | mask=0.1595 | rpn=0.0031 (21.5s)
[ 51/173] total=0.2828 | cls=0.0247 | box=0.0743 | mask=0.1697 | rpn=0.0035 (23.7s)
[ 56/173] total=0.3810 | cls=0.0859 | box=0.0789 | mask=0.2020 | rpn=0.0063 (25.9s)
[ 61/173] total=0.3403 | cls=0.0629 | box=0.0909 | mask=0.1742 | rpn=0.0045 (28.2s)
[ 66/173] total=1.0559 | cls=0.0554 | box=0.0741 | mask=0.9150 | rpn=0.0026 (30.4s)
[ 71/173] total=0.8379 | cls=0.2141 | box=0.1562 | mask=0.4324 | rpn=0.0195 (32.6s)
[ 76/173] total=0.6636 | cls=0.1438 | box=0.1108 | mask=0.3667 | rpn=0.0290 (34.8s)
[ 81/173] total=0.3443 | cls=0.0573 | box=0.0692 | mask=0.1908 | rpn=0.0063 (36.9s)
[ 86/173] total=0.2841 | cls=0.0323 | box=0.0434 | mask=0.1827 | rpn=0.0093 (39.1s)
[ 91/173] total=0.4089 | cls=0.0229 | box=0.0499 | mask=0.3125 | rpn=0.0062 (41.2s)
[ 96/173] total=0.3166 | cls=0.0382 | box=0.0501 | mask=0.2109 | rpn=0.0045 (43.4s)
[101/173] total=0.3529 | cls=0.0322 | box=0.0621 | mask=0.2420 | rpn=0.0067 (45.5s)
[106/173] total=0.1911 | cls=0.0448 | box=0.0462 | mask=0.0880 | rpn=0.0043 (47.6s)
[111/173] total=0.2622 | cls=0.0455 | box=0.0596 | mask=0.1497 | rpn=0.0011 (49.7s)
[116/173] total=0.2385 | cls=0.0341 | box=0.0512 | mask=0.1434 | rpn=0.0014 (51.9s)
[121/173] total=0.5375 | cls=0.0868 | box=0.0858 | mask=0.3507 | rpn=0.0073 (54.0s)
[126/173] total=0.2613 | cls=0.0263 | box=0.0451 | mask=0.1771 | rpn=0.0033 (56.2s)
[131/173] total=0.2105 | cls=0.0405 | box=0.0360 | mask=0.1211 | rpn=0.0042 (58.3s)
[136/173] total=0.1862 | cls=0.0389 | box=0.0488 | mask=0.0797 | rpn=0.0112 (60.4s)
[141/173] total=0.4484 | cls=0.1085 | box=0.0597 | mask=0.2690 | rpn=0.0046 (62.5s)
[146/173] total=0.1954 | cls=0.0313 | box=0.0405 | mask=0.1018 | rpn=0.0149 (64.7s)
[151/173] total=0.3148 | cls=0.0199 | box=0.0243 | mask=0.2615 | rpn=0.0034 (66.8s)
[156/173] total=0.2513 | cls=0.0293 | box=0.0338 | mask=0.1771 | rpn=0.0042 (69.0s)
[161/173] total=0.2460 | cls=0.0293 | box=0.0350 | mask=0.1717 | rpn=0.0023 (71.1s)
[166/173] total=0.2613 | cls=0.0327 | box=0.0379 | mask=0.1810 | rpn=0.0014 (73.3s)
[171/173] total=0.2087 | cls=0.0247 | box=0.0304 | mask=0.1410 | rpn=0.0052 (75.4s)
[173/173] total=0.2086 | cls=0.0156 | box=0.0318 | mask=0.1518 | rpn=0.0021 (76.1s)
```

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EPOCH 2/10  –  TRAINING
[  1/173] total=0.2798 | cls=0.0760 | box=0.0605 | mask=0.1305 | rpn=0.0024 (0.7s)
[  6/173] total=0.1400 | cls=0.0163 | box=0.0174 | mask=0.0965 | rpn=0.0026 (2.8s)
[ 11/173] total=0.1746 | cls=0.0210 | box=0.0285 | mask=0.1120 | rpn=0.0029 (5.0s)
[ 16/173] total=0.1470 | cls=0.0248 | box=0.0327 | mask=0.0813 | rpn=0.0017 (7.2s)
[ 21/173] total=0.7784 | cls=0.0685 | box=0.0573 | mask=0.6422 | rpn=0.0041 (9.4s)
[ 26/173] total=0.2619 | cls=0.0189 | box=0.0304 | mask=0.2001 | rpn=0.0024 (11.6s)
[ 31/173] total=0.2152 | cls=0.0334 | box=0.0377 | mask=0.1350 | rpn=0.0022 (13.9s)
[ 36/173] total=0.2157 | cls=0.0262 | box=0.0375 | mask=0.1419 | rpn=0.0017 (16.1s)
[ 41/173] total=0.2318 | cls=0.0576 | box=0.0337 | mask=0.1312 | rpn=0.0008 (18.3s)
[ 46/173] total=1.2785 | cls=0.0656 | box=0.0625 | mask=1.1421 | rpn=0.0019 (20.5s)
[ 51/173] total=0.1970 | cls=0.0342 | box=0.0301 | mask=0.1236 | rpn=0.0006 (22.8s)
[ 56/173] total=0.1958 | cls=0.0255 | box=0.0225 | mask=0.1379 | rpn=0.0018 (25.0s)
[ 61/173] total=0.1887 | cls=0.0302 | box=0.0394 | mask=0.1056 | rpn=0.0077 (27.2s)
[ 66/173] total=0.2078 | cls=0.0242 | box=0.0158 | mask=0.1590 | rpn=0.0012 (29.5s)
[ 71/173] total=0.4557 | cls=0.0508 | box=0.0452 | mask=0.3443 | rpn=0.0061 (31.7s)
[ 76/173] total=0.1370 | cls=0.0347 | box=0.0175 | mask=0.0749 | rpn=0.0014 (33.8s)
[ 81/173] total=0.1512 | cls=0.0274 | box=0.0212 | mask=0.0949 | rpn=0.0013 (36.1s)
[ 86/173] total=0.2963 | cls=0.0574 | box=0.0443 | mask=0.1895 | rpn=0.0008 (38.2s)
[ 91/173] total=0.2290 | cls=0.0625 | box=0.0441 | mask=0.1144 | rpn=0.0025 (40.4s)
[ 96/173] total=0.2280 | cls=0.0350 | box=0.0295 | mask=0.1531 | rpn=0.0037 (42.6s)
[101/173] total=0.1308 | cls=0.0353 | box=0.0252 | mask=0.0585 | rpn=0.0023 (44.8s)
[106/173] total=0.1679 | cls=0.0253 | box=0.0307 | mask=0.1021 | rpn=0.0015 (46.9s)
[111/173] total=0.1784 | cls=0.0260 | box=0.0217 | mask=0.1214 | rpn=0.0018 (49.1s)
[116/173] total=0.1764 | cls=0.0174 | box=0.0176 | mask=0.1358 | rpn=0.0012 (51.3s)
[121/173] total=0.1644 | cls=0.0252 | box=0.0204 | mask=0.1090 | rpn=0.0039 (53.4s)
[126/173] total=0.2526 | cls=0.0523 | box=0.0292 | mask=0.1563 | rpn=0.0083 (55.6s)
[131/173] total=0.3173 | cls=0.0574 | box=0.0456 | mask=0.2040 | rpn=0.0041 (57.7s)
[136/173] total=0.1659 | cls=0.0240 | box=0.0178 | mask=0.1154 | rpn=0.0036 (59.9s)
[141/173] total=0.6686 | cls=0.1493 | box=0.1297 | mask=0.3568 | rpn=0.0243 (62.1s)
[146/173] total=0.1579 | cls=0.0270 | box=0.0184 | mask=0.0969 | rpn=0.0084 (64.2s)
[151/173] total=0.1973 | cls=0.0210 | box=0.0215 | mask=0.1447 | rpn=0.0037 (66.4s)
[156/173] total=0.1501 | cls=0.0205 | box=0.0357 | mask=0.0836 | rpn=0.0020 (68.5s)
[161/173] total=0.2368 | cls=0.0172 | box=0.0174 | mask=0.1928 | rpn=0.0015 (70.7s)
[166/173] total=0.2592 | cls=0.0213 | box=0.0295 | mask=0.1977 | rpn=0.0017 (72.9s)
[171/173] total=0.3100 | cls=0.0165 | box=0.0148 | mask=0.2652 | rpn=0.0081 (75.1s)
[173/173] total=0.1970 | cls=0.0182 | box=0.0139 | mask=0.1577 | rpn=0.0021 (75.7s)
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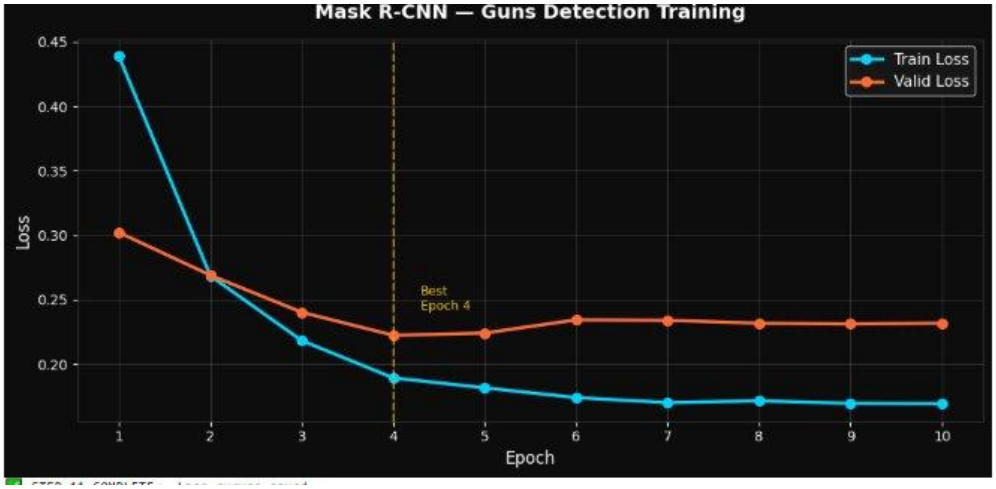
EPOCH 10/10  –  TRAINING
  1/173] total=0.1073 | cls=0.0134 | box=0.0099 | mask=0.0774 | rpn=0.0006 (0.6s)
  6/173] total=0.1618 | cls=0.0396 | box=0.0104 | mask=0.1062 | rpn=0.0002 (2.8s)
 11/173] total=0.2400 | cls=0.0511 | box=0.0112 | mask=0.1717 | rpn=0.0003 (5.0s)
 16/173] total=0.1149 | cls=0.0194 | box=0.0093 | mask=0.0794 | rpn=0.0019 (7.2s)
 21/173] total=0.1323 | cls=0.0103 | box=0.0062 | mask=0.1082 | rpn=0.0011 (9.4s)
 26/173] total=0.1165 | cls=0.0103 | box=0.0093 | mask=0.0918 | rpn=0.0002 (11.5s)
 31/173] total=0.1120 | cls=0.0132 | box=0.0109 | mask=0.0818 | rpn=0.0002 (13.7s)
 36/173] total=0.1930 | cls=0.0454 | box=0.0245 | mask=0.1175 | rpn=0.0004 (15.9s)
 41/173] total=0.1073 | cls=0.0117 | box=0.0111 | mask=0.0781 | rpn=0.0004 (18.1s)
 46/173] total=0.1742 | cls=0.0073 | box=0.0081 | mask=0.1487 | rpn=0.0046 (20.3s)
 51/173] total=0.1154 | cls=0.0151 | box=0.0072 | mask=0.0883 | rpn=0.0003 (22.4s)
 56/173] total=0.1048 | cls=0.0097 | box=0.0094 | mask=0.0794 | rpn=0.0005 (24.6s)
 61/173] total=0.2773 | cls=0.0568 | box=0.0544 | mask=0.1546 | rpn=0.0039 (26.8s)
 66/173] total=0.1441 | cls=0.0097 | box=0.0064 | mask=0.1232 | rpn=0.0004 (29.0s)
 71/173] total=0.2127 | cls=0.0219 | box=0.0415 | mask=0.1443 | rpn=0.0004 (31.2s)
 76/173] total=0.1159 | cls=0.0103 | box=0.0078 | mask=0.0935 | rpn=0.0005 (33.4s)
 81/173] total=0.1310 | cls=0.0364 | box=0.0071 | mask=0.0802 | rpn=0.0007 (35.5s)
 86/173] total=0.1240 | cls=0.0166 | box=0.0139 | mask=0.0891 | rpn=0.0006 (37.7s)
 91/173] total=0.1262 | cls=0.0152 | box=0.0075 | mask=0.0970 | rpn=0.0017 (39.9s)
 96/173] total=0.1112 | cls=0.0096 | box=0.0110 | mask=0.0834 | rpn=0.0012 (42.0s)
101/173] total=0.1325 | cls=0.0160 | box=0.0228 | mask=0.0869 | rpn=0.0014 (44.2s)
106/173] total=0.1899 | cls=0.0511 | box=0.0169 | mask=0.1148 | rpn=0.0003 (46.4s)
111/173] total=0.1185 | cls=0.0096 | box=0.0110 | mask=0.0923 | rpn=0.0013 (48.6s)
116/173] total=0.0875 | cls=0.0115 | box=0.0087 | mask=0.0609 | rpn=0.0011 (50.8s)
121/173] total=0.1676 | cls=0.0458 | box=0.0204 | mask=0.0959 | rpn=0.0004 (52.9s)
126/173] total=0.1764 | cls=0.0430 | box=0.0246 | mask=0.0998 | rpn=0.0034 (55.1s)
131/173] total=0.0832 | cls=0.0120 | box=0.0081 | mask=0.0580 | rpn=0.0001 (57.3s)
136/173] total=0.0856 | cls=0.0115 | box=0.0087 | mask=0.0593 | rpn=0.0004 (59.5s)
141/173] total=0.2471 | cls=0.0818 | box=0.0407 | mask=0.1191 | rpn=0.0008 (61.6s)
146/173] total=0.1237 | cls=0.0142 | box=0.0088 | mask=0.0947 | rpn=0.0004 (63.8s)
151/173] total=0.1215 | cls=0.0115 | box=0.0076 | mask=0.0976 | rpn=0.0001 (66.0s)
156/173] total=0.1756 | cls=0.0456 | box=0.0221 | mask=0.1021 | rpn=0.0005 (68.1s)
161/173] total=0.1211 | cls=0.0100 | box=0.0097 | mask=0.0915 | rpn=0.0042 (70.3s)
166/173] total=0.1348 | cls=0.0118 | box=0.0092 | mask=0.1101 | rpn=0.0003 (72.5s)
171/173] total=0.0831 | cls=0.0127 | box=0.0060 | mask=0.0593 | rpn=0.0001 (74.6s)
173/173] total=0.0734 | cls=0.0081 | box=0.0037 | mask=0.0585 | rpn=0.0001 (75.3s)

Epoch 10 TRAIN avg loss: 0.16951

EPOCH 10  –  VALIDATION
Epoch 10 VALID avg loss: 0.23194
```

Task 4 — Loss Curves and Analysis



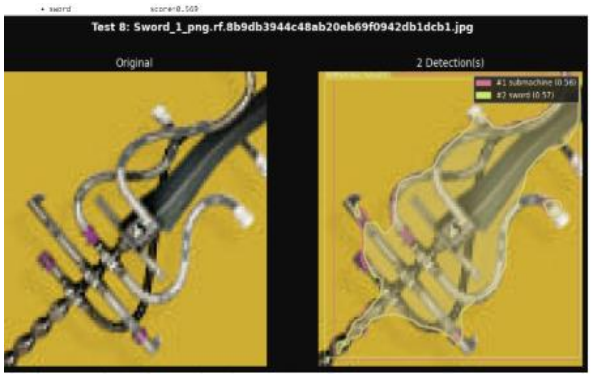
Observations

- Training loss dropped sharply from 0.439 (Epoch 1) to 0.170 (Epoch 10) , consistent learning throughout.
- Validation loss improved from 0.302 to 0.222 by Epoch 4, then plateaued and slightly increased beyond Epoch 5.
- Best model was saved at Epoch 4 (val_loss = 0.22251), marked by the dashed yellow line.
- Learning rate decayed at Epoch 3 (0.005 to 0.0005) and Epoch 6 (to 0.00005) via StepLR, visible as slower training loss descent.
- The gap between training and validation loss from Epoch 5 onward indicates mild overfitting on the small dataset (345 training images).
- Dominant loss component throughout was the mask loss, reflecting the difficulty of pixel-level segmentation on varied weapon shapes.

Task 5: Results

Test Set Summary

- Test images : 15
- Total dets : 16
- Avg/image : 1.1
- Best model : Epoch 4 (val_loss = 0.22251)



Results and Analysis

- The model correctly identified submachine guns across all displayed test images with confidence scores ranging from 0.74 to 0.95.
- High confidence detections (>0.90) show clean, well-fitted masks around the weapon outline.
- Lower confidence detections (0.74) still produce correct class labels but the mask boundary is less precise, likely due to partial occlusion or unusual viewing angle.
- The model generalised well despite the class imbalance, submachine had 228 instances vs. classes like warlock (3) and auto_riff (3) which were likely not detected in the small 15-image test set.
- Average of 1.1 detections per image is consistent with the single-object-per-image nature of the dataset.

Output Files

```
STATS:
Classes     : ['Guns', 'auto_riff', 'snipper', 'submachine', 'sword', 'titan', 'trace riffle', 'warlock']
Best val loss: 0.22251
Test images  : 15
Total dets   : 16
Avg/image    : 1.1
```

Conclusion

Mask R-CNN was successfully trained on a custom 8-class weapons dataset using PyTorch. The model achieved a best validation loss of 0.22251 at Epoch 4 and produced confident, pixel-level instance segmentation masks on unseen test images. Training and validation loss both decreased sharply in the first 4 epochs before the validation curve plateaued, indicating the model was well-fitted within the available data. Given the small dataset size, the results are strong, particularly for the dominant submachine class. Further improvement could be achieved by expanding the dataset for underrepresented classes such as warlock, auto_riff, and Guns.